RK2206实训箱操作手册—SHT40开发

VEML7700光照传感器

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2. 实验内容

在RK2206实训箱上使用I2C接口，实现SHT40温湿度传感器的编程开发。

3. SHT40 API说明

I2C相关API详见PPT。

3.1 函数说明

3.1.1 (一)、SHT40 初始化

定义:	**sht40_init(void)**
功能:	**初始化 SHT40 温湿度传感器模块**
参数:	**void**
返回值:	**void**
依赖:	**\code_base\b5_i2c_sht40\include\sht40.h**

定义:	**sht40_init(void)**
Tips:	**该接口只需调用一次，不支持重复调用。内部会初始化 I2C0_M2 端口，频率为 100KHz**

Table 1

3.1.2 (二)、SHT40 读取温湿度

定义:	**sht40_read_data(sht40_data_ptr dat)**
功能:	**读取 SHT40 的温度与相对湿度**
参数:	`dat`**: 指向存储温度 (°C) 与湿度 (%RH) 的结构体指针**
返回值:	**void**
依赖:	**\code_base\b5_i2c_sht40\include\sht40.h**
Tips:	**内部包含 CRC 校验；温度、湿度会按数据手册公式转换为物理量**

Table 2

3.2 常量说明

3.2.1 I2C 端口

- `EI2C0_M2`

3.2.2 I2C 地址:

- `0x44`

4. 代码编译

4.1 添加文件

将`~vendor/company_demo/b5_i2c_sht40`这整个文件夹拖入`~/application/samples`路径下

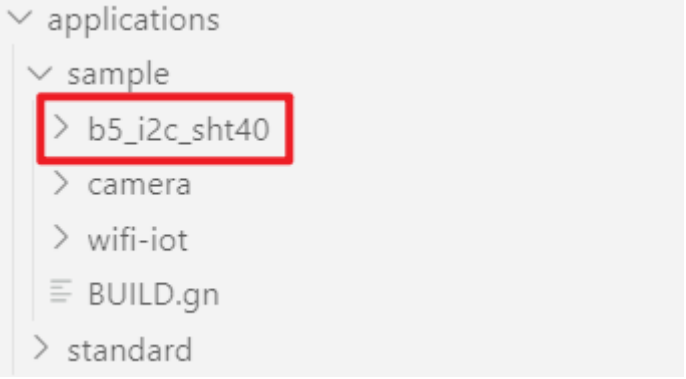


Figure 1

4.2 修改 BUILD.gn 文件

修改 `~/application/samples/` 路径下 BUILD.gn 文件，将如下代码添加到对应位置，指定 `sht40_example` 参与编译。

```
"./b5_i2c_sht40:sht40_example",
```

Fence 1

```
import("//build/lite/config/component/lite_component.gni")

lite_component("app") {
    features = [
        "./b5_i2c_sht40:sht40_example",
    ]

    deps = [
    ]
}
```

Figure 2

修改 `device/lockzhiner/rk2206/sdk_liteos` 路径下 Makefile 文件，添加 `-lveml7700_example` 参与编译。

```
hardware_LIBS = -lhal_iothardware -lhardware -lshellcmd -lsht40_example
```

Fence 2

```
#######################################
# LDFLAGS
#######################################
LDSCRIPT = board.ld

boot_LIBS = -lbootstrap -lbroadcast
hardware_LIBS = -lhal_iothardware -lhardware -lshellcmd -lsht40_example
hdf_LIBS = -lhdf_config -lhdf_core -lhdf_osal_lite -lhdf_platform_lite \
#-lsample_driver -lgpio_driver -lpwm_driver -li2c_driver -ladc_driver -lspi_driver
```

Figure 3

4.3 编译结果

```
root@05fdda9d5ead: /home/
[OHOS INFO] [829/851] STAMP obj/kernel/liteos_m/components/fs/fs.stamp
[OHOS INFO] [830/851] AR libs/libshell.a
[OHOS INFO] [831/851] AR libs/libhichainsdk.a
[OHOS INFO] [832/851] STAMP obj/base/security/deviceauth/frameworks/deviceauth_lite/hichainsdk.stamp
[OHOS INFO] [833/851] gcc cross compiler obj/third_party/paho_mqtt/MQTTClient-C/src/libpahomqtt_static.MQTTClient.o
[OHOS INFO] [834/851] AR libs/libpahomqtt_static.a
[OHOS INFO] [835/851] AR libs/liblwip.a
[OHOS INFO] [836/851] STAMP obj/kernel/liteos_m/kernel.stamp
[OHOS INFO] [837/851] ACTION //test/xts/acts/build_lite:acts_xdevice(//build/lite/toolchain:arm-none-eabi-gcc)
[OHOS INFO] [838/851] STAMP obj/test/xts/acts/build_lite/acts_xdevice.stamp
[OHOS INFO] [839/851] ACTION //test/xts/acts/build_lite:acts_generate_module_data(//build/lite/toolchain:arm-none-eabi-g
cc)
[OHOS INFO] [840/851] STAMP obj/test/xts/acts/build_lite/acts_generate_module_data.stamp
[OHOS INFO] [841/851] ACTION //vendor/lockzhiner/rk2206/hdf_config:hc_gen(//build/lite/toolchain:arm-none-eabi-gcc)
[OHOS INFO] [842/851] STAMP obj/vendor/lockzhiner/rk2206/hdf_config/hc_gen.stamp
[OHOS INFO] [843/851] gcc cross compiler obj/out/rk2206/lockzhiner-rk2206/gen/vendor/lockzhiner/rk2206/hdf_config/libhdf
_config.hdf_hex.o
[OHOS INFO] [844/851] AR libs/libhdf_config.a
[OHOS INFO] [845/851] STAMP obj/device/rockchip/rk2206/sdk_liteos/sdk.stamp
[OHOS INFO] [846/851] STAMP obj/device/rockchip/rk2206/sdk_liteos/wifiiot_sdk.stamp
[OHOS INFO] [847/851] ACTION //test/xts/acts/build_lite:acts(//build/lite/toolchain:arm-none-eabi-gcc)
[OHOS INFO] [848/851] STAMP obj/test/xts/acts/build_lite/acts.stamp
[OHOS INFO] [849/851] STAMP obj/build/lite/ohos.stamp
[OHOS INFO] [850/851] ACTION //device/rockchip/rk2206/sdk_liteos:liteos(//build/lite/toolchain:arm-none-eabi-gcc)
[OHOS INFO] [851/851] STAMP obj/device/rockchip/rk2206/sdk_liteos/liteos.stamp
[OHOS INFO] /home/openharmony/vendor/lockzhiner/rk2206/fs.yml not found, stop packing fs. If the product does not need t
o be packaged, ignore it.
[OHOS INFO] lockzhiner-rk2206 build success
[OHOS INFO] cost time: 0:01:34
root@05fdda9d5ead:/home/openharmony#
```

Figure 4

4.4 烧录固件

打开烧录工具RKDevTool

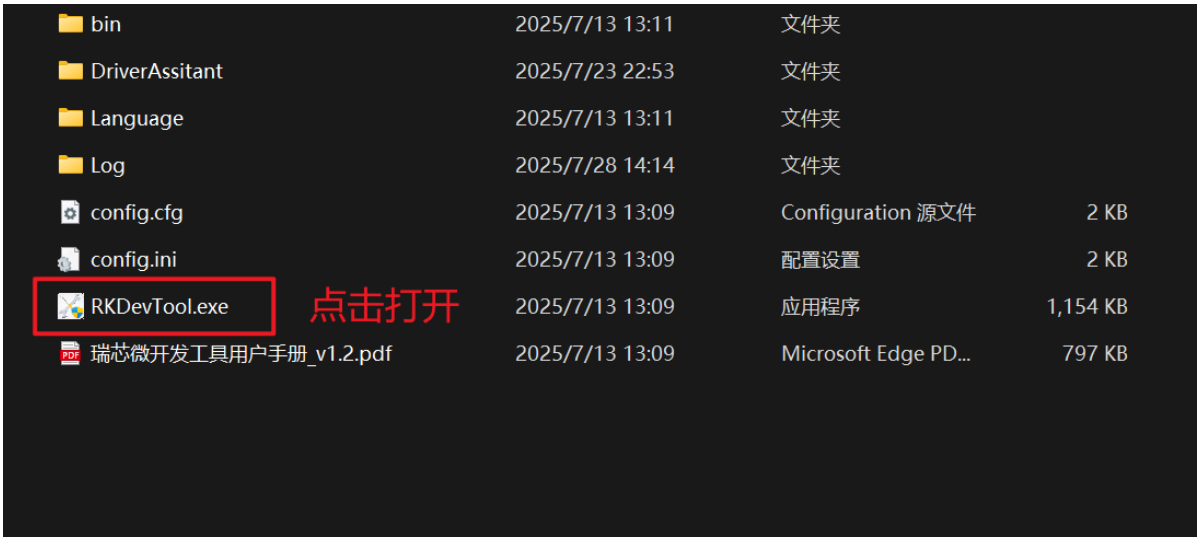
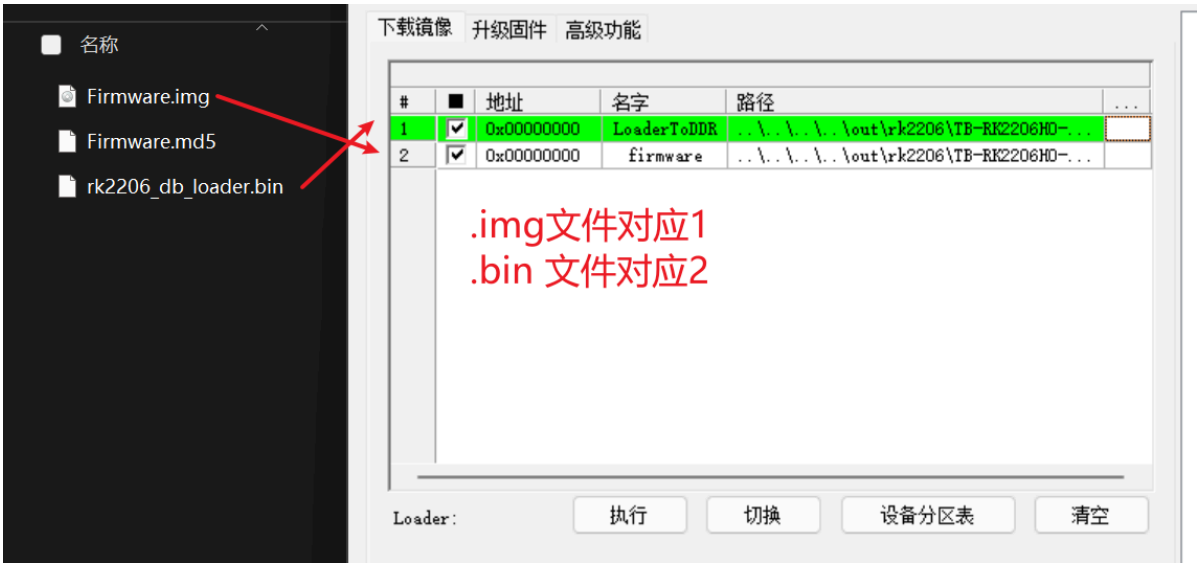


Figure 5

指定两个烧录文件，编译出的目标文件在 `out/rk2206/lockzhiner-rk2206/images` 目录下。



在RK2206实训箱上上，长按左侧模块控制终端的MaskRom按键（MaskRom按键不要松开），点击ReSet按键（短按，按下后松开，但MaskRom按键不要松开，大约5秒后松开)，烧写工具出现："发现一个MASKROM设备"。如下图所示：

Figure 6

此时点击"执行"，下载程序，下载完成如下：

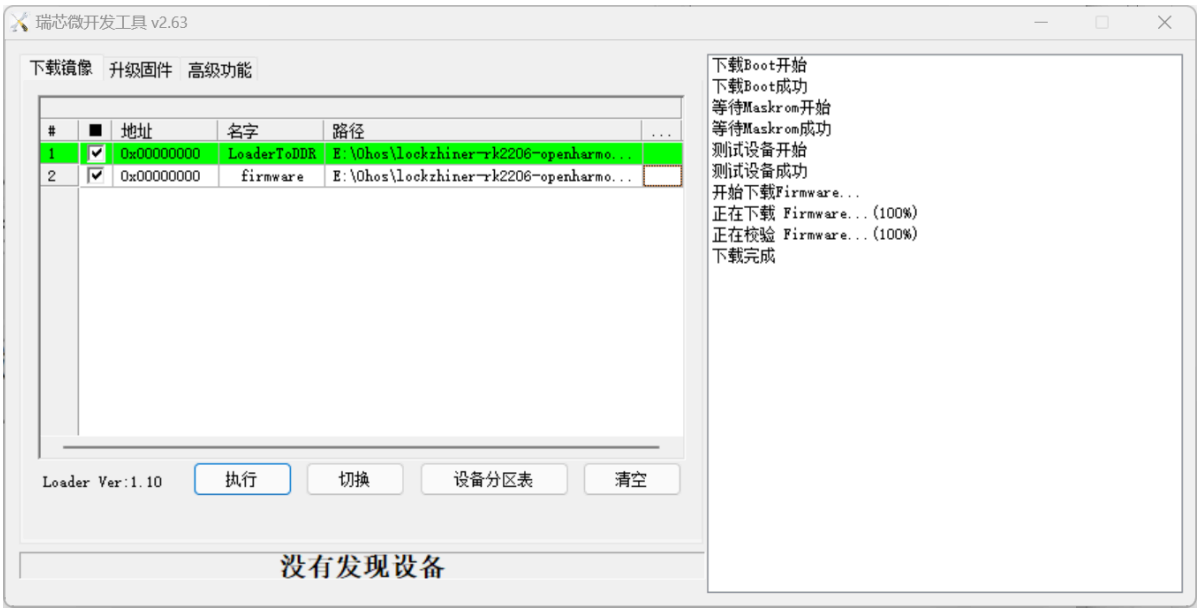


Figure 7

4.5 运行结果

示例代码编译烧录代码后，按下开发板的RESET按键，连接串口，可以看到温湿度信息。

```
entering kernel init...
hilog will init.
[MAIN:D]Main: LOS_Start ...
Entering scheduler
OHOS # hiview init success.
[GPIO:D]LzGpioInit: id 1 is initialized successfully
[GPIO:D]LzGpioInit: id 0 is initialized successfully
temperature: 28.43, humidity: 41.67
temperature: 28.44, humidity: 41.66
temperature: 28.46, humidity: 41.68
temperature: 28.47, humidity: 41.67
temperature: 28.44, humidity: 41.69
temperature: 28.48, humidity: 41.66
temperature: 28.46, humidity: 41.66
temperature: 28.46, humidity: 41.67
temperature: 28.44, humidity: 41.68
temperature: 28.43, humidity: 41.69
temperature: 28.41, humidity: 41.65
temperature: 28.46, humidity: 41.64
temperature: 28.44, humidity: 41.63
temperature: 28.43, humidity: 41.61
temperature: 28.44, humidity: 41.64
temperature: 28.43, humidity: 41.59
temperature: 28.43, humidity: 41.63
temperature: 28.43, humidity: 41.64
temperature: 28.42, humidity: 41.62
temperature: 28.44, humidity: 41.64
```

Fence 3

```
entering kernel init...
hilog will init.
[MAIN:D]Main: LOS_Start ...
Entering scheduler
OHOS # hiview init success.[GPIO:D]LzGpioInit: id 1 is initialized successfully
[GPIO:D]LzGpioInit: id 0 is initialized successfully
temperature: 26.20, humidity: 47.49
temperature: 26.16, humidity: 47.52
temperature: 26.20, humidity: 47.53
temperature: 26.19, humidity: 47.54
temperature: 26.20, humidity: 47.55
temperature: 26.20, humidity: 47.55
temperature: 26.17, humidity: 47.96
temperature: 26.20, humidity: 48.36
temperature: 26.22, humidity: 48.36
temperature: 26.45, humidity: 48.78
temperature: 26.55, humidity: 53.11
temperature: 26.68, humidity: 57.79
temperature: 26.84, humidity: 61.22
temperature: 27.34, humidity: 63.24
temperature: 27.96, humidity: 63.45
temperature: 28.26, humidity: 63.16
temperature: 27.99, humidity: 60.12
```

Figure 8

5. 注意事项

1. **初始化顺序**: 必须先调用 `sht40_init()` 再进行读取
2. **I2C 通信**: 设备地址为 `0x44`，命令触发后需等待约 10ms 再读数据
3. **CRC 校验**: 驱动已自动进行 CRC 校验，校验失败会跳过对应物理量更新
4. **数据单位**: 温度单位为 °C，湿度单位为 %RH

6. 技术规格

- **通信接口**: I2C
- **I2C 地址**: 0x44
- **输出**: 温度与相对湿度
- **分辨率**: 16 位原始数据（内部转换）
- **典型测量等待**: 10 ms